

Birla Institute of Technology and Science, Pilani

Work Integrated Learning Programmes Division

M.Tech. in Artificial Intelligence and Machine Learning

II Semester 2025-26

Course Number	AIML ZC416	
Course Name	Mathematical Foundation for Machine Learning	
Nature of Exam	Closed book	# Pages 2
Weightage for grading	30%	# Questions 5
Duration	120 minutes	
Date of Exam	15-03-2026 (FN)	

Instructions

1. All questions are compulsory
 2. State your assumptions clearly wherever required. No doubts will be entertained in the examination hall.
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Q1 Given an input $[x, y]^T$, an output $f(x, y) = x + y + \cos(x) + \cos(y)$ is generated by a process.

- a) Using an appropriate method, find a linear model $L(x, y)$ to predict the output of the given process about $(0, 0)$. [1 Mark]
- b) If the loss function is defined as $h(x, y) = (f(x, y) - L(x, y))^2$ then find the gradient and the Hessian of $h(x, y)$. [2 Marks]
- c) Find all critical points of $h(x, y)$ and classify them [2 Marks]

Q2 a) Find all the solutions of the following system. [2 Marks]

$$\begin{aligned}x_1 + 7x_3 + 14x_4 &= 40 \\ 3x_1 + 4x_2 + 2x_3 + 5x_4 &= 6\end{aligned}$$

- b) Let $V = \mathbb{R}^2$ and define the inner operation $\mathbf{x} \oplus \mathbf{y} = [x_1 + y_1 + 1, x_2 + y_2 + 1]^T$, where $\mathbf{x} = [x_1, x_2]^T, \mathbf{y} = [y_1, y_2]^T \in V$, and define the outer operation as $k[x_1, x_2]^T = [kx_1, kx_2]^T$ where $k \in \mathbb{R}$. Prove or disprove that V is a vector space over $\mathbb{R}$ with the given operations [2 Marks]
- c) Let $B = \{b_1, b_2\}$ be the basis of a two-dimensional subspace $U \subseteq \mathbb{R}^3$ where

$$b_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \text{ and } b_2 = \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix}$$

Using an appropriate method, convert B into an orthonormal basis $C = \{c_1, c_2\}$ of U . [2 Marks]

- d) Find matrices A, B such that A^{-1}, B^{-1} and $(A + B)^{-1}$ exist but $(A + B)^{-1} \neq A^{-1} + B^{-1}$. [1 Mark]

Q3 A machine learning engineer is developing a multivariate Gaussian generative model for a dataset with three correlated features. The covariance matrix and mean vector are given by

$$\Sigma = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 5 & 1 \\ 2 & 1 & 3 \end{bmatrix}, \quad \mu = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}.$$

- (a) Prove that Σ is symmetric, positive definite. [2 Marks]
 (b) Compute the Cholesky decomposition $\Sigma = L L^T$. Explain why the condition in (a) is necessary for Cholesky decomposition. [4 Marks]

Q4 Answer the following with justifications.

- a) Is it in general true that a non-zero eigenvalue of $\mathbf{A}^T \mathbf{A}$ is an eigenvalue of $\mathbf{A} \mathbf{A}^T$ and vice versa? What can be said about the eigenvectors? [2 Marks]
 b) Given a non-zero vector $\mathbf{r} \in \mathbb{R}^k$, what can be said about the eigenvalues and eigenvectors of $\mathbf{r} \mathbf{r}^T$? [1 Mark]
 c) Is it in general true that the rank of a matrix is equal to the number of non-zero eigenvalues? [1 Mark]
 d) Can the eigenvalue decomposition and SVD for a matrix be the same? Give the characterization, if possible. [1 Mark]
 e) Is it true that the eigenvalues of a lower triangular matrix are its diagonal entries? [1 Mark]

Q5 Answer the following questions.

- a) Without direct multiplication, write a pseudocode to find out if $\mathbf{U}, \Sigma$ and $\mathbf{V}$ are the matrices associated with the SVD of a matrix $\mathbf{A}$ of size $m \times n$. You may use the function $[\mathbf{E}, \mathbf{W}] = \text{eigs}(\mathbf{M})$, that stores the eigenvalues of $\mathbf{M}$ in $\mathbf{W}$ and the corresponding eigenvectors in $\mathbf{E}$. State your assumptions, if any, explicitly. [2 Marks]
 b) Write a pseudocode to convert a square matrix $\mathbf{A}$ of size $n \times n$ in upper triangular form to RREF assuming that the determinant of $\mathbf{A}$ is non-zero. [2 Marks]
 c) Write a pseudocode to get the inner product of two vectors $\mathbf{a}$ and $\mathbf{b}$ with respect to the basis $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$. Assume that the symmetric, positive definite matrix required to compute the inner product is $\mathbf{A}$ and $\text{inv}(\mathbf{X})$ gives the inverse of a matrix $\mathbf{X}$. [2 Marks]